UNNS-U2-TI

Target-Blind Selector Language and Generative Knowledge Accounting

UNNS Research Collective

Version v001 (addendum to UNNS-U2-KI v001, which is frozen; R3, FP-SYM and earlier releases are unchanged)

Abstract

KI v001 measured how much mathematical knowledge three route selectors use. This addendum asks the converse: what do fixed, sequence-independent, graph-local rules produce on their own? **TI-0.** The exact-arithmetic gate passes: all 46 route-label expressions have integer coefficients and no executed selector uses floating point. This corrects a precaution stated in KI v001. **TI-1.** We freeze a grammar GL-1 with 360 finite-control rules (class F, 5040 orbits over 14 starts) plus a reference class M with unbounded memory (120 rules), and audit that no rule reads a target, step counter or table. **TI-2.** In the registered run of 6720 orbits, 1373 orbits meet the registered success criterion, but they come from only 56 distinct prefixes and are geometric growth ($\times 4$ or $\times 25$ per two steps). No orbit reproduces Fibonacci numbers, consecutive squares, primes, triangular numbers or consecutive members of any registered family. **TI-3.** Under a fixed encoding, a graph-local rule costs 75 bytes plus a 4103-byte interpreter and no sequence-specific table; the knowledge-rich selectors use $\geq$870 compressed table bytes. These numbers describe different tasks and are not a complexity comparison. Target-independent generation of a nontrivial sequence is *not* shown.

1 Scope and evidence levels

Certified exactly: TI-0 (integer polynomial coefficients, exact comparison against the frozen evaluator). *Tested*: the registered run, the audits, the accounting. *Exploratory*: analyses added after results were seen (labelled where they occur). "No hit in the registered vocabulary" is never read as impossibility.

2 TI-0: exact-arithmetic gate

All 46 label expressions of the three KI selectors were evaluated as integer Horner forms from the stored coefficient polynomials. There are zero non-integer coefficients and zero labels with division or float literals. A static AST scan of the executed selector sources finds only true-division nodes in SymPy symbolic code and Fraction arithmetic. On 5435 inputs (squares n^2, n up to 10^{15}) the exact evaluator, the KI shortcut evaluator and the frozen lambdified evaluator agree on every move sequence and final state; on 150 random $n \in [10^{10}, 10^{60})$ they also agree. A float trap confirms the exact path raises on any float.

Correction. KI v001 listed floating-point evaluation of the U2-001 labels as a caveat. TI-0 shows the labels are exact integers, so the caveat was a precaution that did not materialize. KI v001 stays frozen; this paragraph is the correction.

3 TI-1: the language GL-1 and target-blindness

A rule is a tuple (dir, hist, band, key, order). At vertex x it enumerates graph neighbours y (from divisors and square divisors of x), filters by direction, by the band ($y \le 2x$, $y \ge x/2$, or both) and by the last ≤ 2 vertices (class F) or the whole visited set (class M), then picks the min or max of a key (y, N, $y \bmod 4$, $y \bmod 6$, $v_2(y)$), ties by smallest y. Information channels are X (current vertex), H (window ≤ 2), NB (neighbours), and for M only V (visited set). Horizon $T = 400$ and escape bound 10^{18} act outside the selector.

Target-blindness conditions. TB1 no target or table is an input; TB2 no step counter; TB3 the same rule is applied to every start; TB4 class F reads only X, H, NB; TB5 the interpreter holds no sequence constants. *Audit* (verified by script): neighbour sets agree with an independent trial-division engine for $x \le 3000$ (0 mismatches); 14774 class-F states re-executed from (x, hist) alone reproduce the step (0 mismatches); appending history beyond the window never changes a step (0 mismatches); class F made 0 reads of V; the interpreter imports only `sys` and the graph library, reads no file, and has integer literals $\{0, 1, 2, 4, 6\}$ only. This audits the declared language, not every conceivable hidden dependency.

Registration. Config, interpreter and recognizers are SHA-256 pinned; the verifier checks the hashes and that the stored results equal a fresh re-run. Recognizers: Berlekamp–Massey over $\mathbb{Q}$ (order ≤ 4) on the first 60 terms, and family tests on naturals, squares, cubes, triangular numbers, 2^k, 3^k, Fibonacci, factorials and primes. **Amendment A1** reduced the BM window from 400 to 60 terms after a timing test of a single rule hung on large integers; this was made before any complete run, and the only endpoints seen were from that one rule.

4 TI-2: what the rules produce

class	cyclic	escape	horizon	stuck
F (360 rules)	152	1410	26	3452
M (120 rules)	0	476	34	1170

(14 starts per rule, 6720 orbits.) 1373 orbits (F 1027, M 346) satisfy the registered success criterion (a linear recurrence of order ≤ 4, or consecutive family members, with no cyclic ending). Their sequences are:

- $1, 4, 16, 64, 256, \ldots$ (move +1 repeated; recurrence $a_n = 4a_{n-1}$);
- $3, 12, 75, 300, 1875, 7500, \ldots$ (alternating +1, +4; $\times 25$ every two steps);
- the same pattern with periodic exceptions (e.g. $4, 18, 147, 588, \ldots$).

All successes are of these forms. **This is trivial geometric growth, not evidence that the rules generate the intended sequences.** No orbit reaches Fibonacci (0 consecutive-Fibonacci hits), and 0 orbits show consecutive members of any registered family. Family "subset" hits (naturals, powers of 2, squares) are weak signals: a "subset" needs only 8 family members among the terms, and no consecutive run was found.

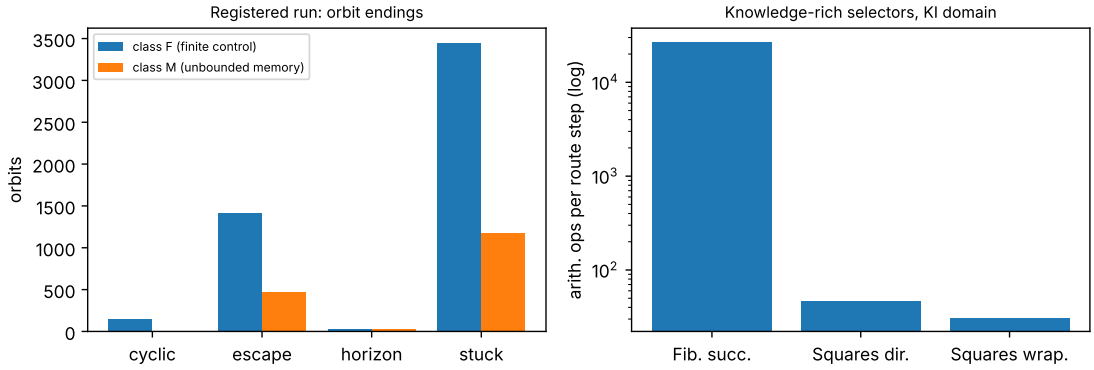
Exploratory (after the registered results). Of 1946 non-cyclic orbits that escape or hit the horizon, 448 follow $x \mapsto (x+1)^2$ (e.g. $1, 4, 25, 676, 458329, \ldots$). These are *not* registered successes because they leave the 10^{18} bound after about 7 terms, so the registered recognizers never see enough terms. The map $x \mapsto (x+1)^2$ is a one-line law; it is not a target sequence.

5 TI-3: knowledge accounting (fixed encoding)

Encoding: canonical minified JSON of the data each selector consults; zlib level 9 as a rough proxy; code bytes = UTF-8 selector source size.

selector	table B	zlib B	code B	states	moves/step	ops/step
Fibonacci successor (KI)	16906	3118	5219	201	8.25	26937
Squares direct (KI)	12093	946	5457	85	4.37	46
Squares wrapper (KI)	7375	870	9765	45	6.92	31
Graph-local rule (class F)	75	64	4103	—	1	84.1 cand.

A class-F rule is $\log_2 360 \approx 8.49$ bits of choice. *Caution:* the knowledge-rich selectors route to a prescribed target (the next Fibonacci number or square); the graph-local rules do not, and they never produce those targets. A smaller table is therefore not evidence of lower intrinsic complexity: the graph-local rules simply solve a different, easier problem. The honest reading is that the information the TI rules lack is exactly what the KI selectors store.



6 The R3′ decision

R3′ (exact integer $+, -, \times$, division by constants, comparison, isqrt, residues) is adopted as a separate, named *algebraic-monotone selector class*. Definition 2.1 of R3 is not modified retroactively.

7 Gates, assumptions, consequences

Gates passed: TI-0 exact (0 mismatches), registration hashes, stored equals fresh (6720 orbits), neighbour/locality/perturbation audits. Unverified: that GL-1 exhausts target-blind finite-control rules (it does not); that other recognizers would find nothing; behaviour beyond $T = 400$ or 10^{18}; whether richer grammars or interpreters generate primes. Consequences: within GL-1 no nontrivial target sequence is generated; "target-independent intrinsic generation" remains open. The unresolved quadratic-tower cases and class-11 optimization are separate follow-ups, not prerequisites.